

## Appendix A: Basic Algorithm Information

| Algorithm Name                                           | Scloud+                                                                                                                                      |                                                       |
|----------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------|
| <b>Algorithm Functionality</b><br><b>(Single choice)</b> | <input type="checkbox"/> Digital Signature<br><input checked="" type="checkbox"/> Key Encapsulation<br><input type="checkbox"/> Key Exchange |                                                       |
| <b>Submitter 1</b><br><br><u>王安宇</u><br><br>2026-04-28   | Name                                                                                                                                         | Anyu Wang                                             |
|                                                          | Affiliation                                                                                                                                  | Tsinghua University                                   |
|                                                          | Phone Number                                                                                                                                 | +8615201173389                                        |
|                                                          | Email                                                                                                                                        | anyuwang@tsinghua.edu.cn                              |
|                                                          | Postal Address                                                                                                                               | Tsinghua University, Beijing, China                   |
| <b>Submitter 2</b><br><br><u>郑中翔</u><br><br>2026-04-28   | Name                                                                                                                                         | Zhongxiang Zheng                                      |
|                                                          | Affiliation                                                                                                                                  | Communication University of China                     |
|                                                          | Phone Number                                                                                                                                 | +8615210588700                                        |
|                                                          | Email                                                                                                                                        | zhengzx@cuc.edu.cn                                    |
|                                                          | Postal Address                                                                                                                               | Communication University of China, Beijing, China     |
| <b>Submitter 3</b><br><br><u>赵春环</u><br><br>2026-04-28   | Name                                                                                                                                         | Chunhuan Zhao                                         |
|                                                          | Affiliation                                                                                                                                  | Huawei Technologies                                   |
|                                                          | Phone Number                                                                                                                                 | +8615201073567                                        |
|                                                          | Email                                                                                                                                        | zhaochunhuan@huawei.com                               |
|                                                          | Postal Address                                                                                                                               | Huawei Beijing Research Institute Q27, Beijing, China |
| <b>Submitter 4</b><br><br><u>邱志远</u>                     | Name                                                                                                                                         | Zhiyuan Qiu                                           |
|                                                          | Affiliation                                                                                                                                  | Shandong Institute of Blockchain                      |
|                                                          | Phone Number                                                                                                                                 | +8615662649250                                        |
|                                                          | Email                                                                                                                                        | qiuzhiyuan@sdibc.cn                                   |

## Appendix B: Intellectual Property Statements

### B.1 Statement by Each Algorithm Submitter

I, Anyu Wang, Tsinghua University, Haidian District, Beijing, China, to the best of my knowledge of my submitted algorithm Scloud+ and its implementations (reference and optimized implementations included), do hereby declare:

1. I commit that my submitted algorithm and its implementations are in compliance with the security requirements, including but not limited to the requirements that the algorithm does not contain any backdoor or defect artificially designed and its implementations do not contain any malicious code. I do hereby commit that to the best of my knowledge, my submitted algorithm and its implementations do not contain any data, information or technology that may infringe upon trade secrets, or may involve confidentiality required by the laws and regulations of China and other countries.

2. I acknowledge and agree that my submitted algorithm and its implementations will be provided to the public for evaluation, analysis and review. I agree that any interested party, including the program organizer, public reviewers, etc., can freely use the algorithm and its implementations for evaluation purpose. I acknowledge that the program organizer will select several algorithms as candidates for the next-round evaluation or the finalists after evaluation and analysis; my submitted algorithm might not be selected as a candidate for the next-round evaluation or the finalist. I acknowledge that I will not receive financial or other compensation from the program organizer for my submitted algorithm.

3. To facilitate the next-generation commercial cryptographic algorithm standardization, I acknowledge and agree that if my submitted algorithm is selected for the finalists, the program organizer may modify the algorithm and its implementations out of security, availability or other considerations. If my submitted algorithm is selected for the finalists and to be standardized, I commit to place no restriction on the worldwide disclosure and free adoption of algorithm and its implementations.

4. I commit that the algorithm and its implementations are my (or my team's) own original work. The implementations of my submitted algorithm ☐do/☒do not adopt code or code libraries requiring authorization by third parties. Except those clearly identified, citations and acknowledgment, all the opinions, words, diagrams and data in the submission are my own (or my team's) original research results, and the submission does not contain any published work from other individual or team.

5. (Please check one of the following two options.)

☒ I do not hold and do not intend to hold any patent or patent application which may cover my submitted algorithm and its implementations, and to the best of my knowledge, no patent or patent application may cover my submitted algorithm and its implementations.

☐ The following patent(s) and patent application(s) may cover my submitted

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### B.1 Statement by Each Algorithm Submitter

I, Zhongxiang Zheng, Communication University of China, Haidian district, Beijing, China, to the best of my knowledge of my submitted algorithm Scloud+ and its implementations (reference and optimized implementations included), do hereby declare:

1. I commit that my submitted algorithm and its implementations are in compliance with the security requirements, including but not limited to the requirements that the algorithm does not contain any backdoor or defect artificially designed and its implementations do not contain any malicious code. I do hereby commit that to the best of my knowledge, my submitted algorithm and its implementations do not contain any data, information or technology that may infringe upon trade secrets, or may involve confidentiality required by the laws and regulations of China and other countries.
2. I acknowledge and agree that my submitted algorithm and its implementations will be provided to the public for evaluation, analysis and review. I agree that any interested party, including the program organizer, public reviewers, etc., can freely use the algorithm and its implementations for evaluation purpose. I acknowledge that the program organizer will select several algorithms as candidates for the next-round evaluation or the finalists after evaluation and analysis; my submitted algorithm might not be selected as a candidate for the next-round evaluation or the finalist. I acknowledge that I will not receive financial or other compensation from the program organizer for my submitted algorithm.
3. To facilitate the next-generation commercial cryptographic algorithm standardization, I acknowledge and agree that if my submitted algorithm is selected for the finalists, the program organizer may modify the algorithm and its implementations out of security, availability or other considerations. If my submitted algorithm is selected for the finalists and to be standardized, I commit to place no restriction on the worldwide disclosure and free adoption of algorithm and its implementations.
4. I commit that the algorithm and its implementations are my (or my team's) own original work. The implementations of my submitted algorithm ☐do/☒do not adopt code or code libraries requiring authorization by third parties. Except those clearly identified, citations and acknowledgment, all the opinions, words, diagrams and data in the submission are my own (or my team's) original research results, and the submission does not contain any published work from other individual or team.
5. (Please check one of the following two options.)

I do not hold and do not intend to hold any patent or patent

application which may cover my submitted algorithm and its implementations, and to the best of my knowledge, no patent or patent application may cover my submitted algorithm and its implementations.  
☐ The following patent(s) and patent application(s) may cover my submitted algorithm and its implementations. As for patent(s) and patent application(s) held by myself, I ☐ have submitted/☐ have not submitted all statements; as for patent(s) and patent application(s) held by others, I ☐ have submitted/☐ have not submitted all statements signed by patent and patent application owner(s).

Table The List of Patent(s) and Patent Application(s)

| No. | Patent/Patent Application No. | Title of Patent/Patent Application | Patent/Patent Application Owner(s) | Whether submit statements signed by each owner           |
|-----|-------------------------------|------------------------------------|------------------------------------|----------------------------------------------------------|
|     |                               |                                    |                                    | <input type="checkbox"/> Yes <input type="checkbox"/> No |

Note: The algorithm submitter(s) shall fill in the table according to his/her/their situation.

I do hereby declare that to the best of my knowledge, I have fully disclosed all patents and patent applications which may cover my submitted algorithm and its implementations. I commit that to the best of my knowledge, my submitted algorithm and its implementations are in compliance with the laws and regulations of China and other countries, and do not infringe upon legitimate rights and interests of any third party. Where there is a dispute concerning patent infringement or violation of laws and regulations, I commit that the algorithm submitter(s) shall take the responsibility of dealing with it and eliminating negative effects as well as bearing corresponding legal liabilities.

I am fully aware that I bear the legal consequences of this statement.

Signed: 郑中翔

Date: 2026.6.26

*(Note: If there are multiple algorithm submitters, the statements shall be signed by each and every one of them.)*

## Appendix B: Intellectual Property Statements

### B.1 Statement by Each Algorithm Submitter

I, Chunhuan Zhao, No. 156, Beiqing Road, Haidian District, Beijing 100095, China, to the best of my knowledge of my submitted algorithm ScIoud+ and its implementations (reference and optimized implementations included), do hereby declare:

1. I commit that my submitted algorithm and its implementations are in compliance with the security requirements, including but not limited to the requirements that the algorithm does not contain any backdoor or defect artificially designed and its implementations do not contain any malicious code. I do hereby commit that to the best of my knowledge, my submitted algorithm and its implementations do not contain any data, information or technology that may infringe upon trade secrets, or may involve confidentiality required by the laws and regulations of China and other countries.

2. I acknowledge and agree that my submitted algorithm and its implementations will be provided to the public for evaluation, analysis and review. I agree that any interested party, including the program organizer, public reviewers, etc., can freely use the algorithm and its implementations for evaluation purpose. I acknowledge that the program organizer will select several algorithms as candidates for the next-round evaluation or the finalists after evaluation and analysis; my submitted algorithm might not be selected as a candidate for the next-round evaluation or the finalist. I acknowledge that I will not receive financial or other compensation from the program organizer for my submitted algorithm.

3. To facilitate the next-generation commercial cryptographic algorithm standardization, I acknowledge and agree that if my submitted algorithm is selected for the finalists, the program organizer may modify the algorithm and its implementations out of security, availability or other considerations. If my submitted algorithm is selected for the finalists and to be standardized, I commit to place no restriction on the worldwide disclosure and free adoption of algorithm and its implementations.

4. I commit that the algorithm and its implementations are my (or my team's) own original work. The implementations of my submitted algorithm ☐do/☐do not adopt code or code libraries requiring authorization by third parties. Except those clearly identified, citations and acknowledgment, all the opinions, words, diagrams and data in the submission are my own (or my team's) original research results, and the submission does not contain any published work from other individual or team.

5. *(Please check one of the following two options.)*

- ☐ I do not hold and do not intend to hold any patent or patent application which may cover my submitted algorithm and its implementations, and to the best of my knowledge, no patent or patent application may cover my submitted algorithm and its implementations.

√ The following patent(s) and patent application(s) may cover my submitted algorithm and its implementations. As for patent(s) and patent application(s) held by myself, I ☐ have submitted/☐ have not submitted all statements; as for patent(s) and patent application(s) held by others, I ☒ have submitted/☐ have not submitted all statements signed by patent and patent application owner(s).

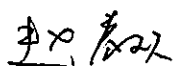
Table The List of Patent(s) and Patent Application(s)

| No. | Patent/Patent Application No. | Title of Patent/Patent Application | Patent/Patent Application Owner(s)               | Whether submit statements signed by each owner                      |
|-----|-------------------------------|------------------------------------|--------------------------------------------------|---------------------------------------------------------------------|
| 1   | 202411158536.6                | 数据编码、解码的方法、装置、芯片和系统                | Huawei Technologies Co., Ltd、Shandong University | <input checked="" type="checkbox"/> Yes <input type="checkbox"/> No |

Note: The algorithm submitter(s) shall fill in the table according to his/her/their situation.

I do hereby declare that to the best of my knowledge, I have fully disclosed all patents and patent applications which may cover my submitted algorithm and its implementations. I commit that to the best of my knowledge, my submitted algorithm and its implementations are in compliance with the laws and regulations of China and other countries, and do not infringe upon legitimate rights and interests of any third party. Where there is a dispute concerning patent infringement or violation of laws and regulations, I commit that the algorithm submitter(s) shall take the responsibility of dealing with it and eliminating negative effects as well as bearing corresponding legal liabilities.

I am fully aware that I bear the legal consequences of this statement.

Signed: 

Date: April 26th, 2026

*(Note: If there are multiple algorithm submitters, the statements shall be signed by each and every one of them.*

## Appendix B: Intellectual Property Statements

### B.1 Statement by Each Algorithm Submitter

I, Zhiyuan Qiu, A7-7, Hanyu Jingu,Jina, China, to the best of my knowledge of my submitted algorithm Scloud+ and its implementations (reference and optimized implementations included), do hereby declare:

1. I commit that my submitted algorithm and its implementations are in compliance with the security requirements, including but not limited to the requirements that the algorithm does not contain any backdoor or defect artificially designed and its implementations do not contain any malicious code. I do hereby commit that to the best of my knowledge, my submitted algorithm and its implementations do not contain any data, information or technology that may infringe upon trade secrets, or may involve confidentiality required by the laws and regulations of China and other countries.

2. I acknowledge and agree that my submitted algorithm and its implementations will be provided to the public for evaluation, analysis and review. I agree that any interested party, including the program organizer, public reviewers, etc., can freely use the algorithm and its implementations for evaluation purpose. I acknowledge that the program organizer will select several algorithms as candidates for the next-round evaluation or the finalists after evaluation and analysis; my submitted algorithm might not be selected as a candidate for the next-round evaluation or the finalist. I acknowledge that I will not receive financial or other compensation from the program organizer for my submitted algorithm.

3. To facilitate the next-generation commercial cryptographic algorithm standardization, I acknowledge and agree that if my submitted algorithm is selected for the finalists, the program organizer may modify the algorithm and its implementations out of security, availability or other considerations. If my submitted algorithm is selected for the finalists and to be standardized, I commit to place no restriction on the worldwide disclosure and free adoption of algorithm and its implementations.

4. I commit that the algorithm and its implementations are my (or my team's) own original work. The implementations of my submitted algorithm ☐do/☒do not adopt code or code libraries requiring authorization by third parties. Except those clearly identified, citations and acknowledgment, all the opinions, words, diagrams and data in the submission are my own (or my team's) original research results, and the submission does not contain any published work from other individual or team.

5. (Please check one of the following two options.)

☒ I do not hold and do not intend to hold any patent or patent application which may cover my submitted algorithm and its implementations, and to the best of my knowledge, no patent or patent application may cover my submitted algorithm and its implementations.

☐ The following patent(s) and patent application(s) may cover my submitted

## B.1 Statement by Each Algorithm Submitter

I, Yi Tang, 156 Beiqing Road, Haidian District, Beijing, China, to the best of my knowledge of my submitted algorithm Scloud+ and its implementations (reference and optimized implementations included), do hereby declare:

1. I commit that my submitted algorithm and its implementations are in compliance with the security requirements, including but not limited to the requirements that the algorithm does not contain any backdoor or defect artificially designed and its implementations do not contain any malicious code. I do hereby commit that to the best of my knowledge, my submitted algorithm and its implementations do not contain any data, information or technology that may infringe upon trade secrets, or may involve confidentiality required by the laws and regulations of China and other countries.

2. I acknowledge and agree that my submitted algorithm and its implementations will be provided to the public for evaluation, analysis and review. I agree that any interested party, including the program organizer, public reviewers, etc., can freely use the algorithm and its implementations for evaluation purpose. I acknowledge that the program organizer will select several algorithms as candidates for the next-round evaluation or the finalists after evaluation and analysis; my submitted algorithm might not be selected as a candidate for the next-round evaluation or the finalist. I acknowledge that I will not receive financial or other compensation from the program organizer for my submitted algorithm.

3. To facilitate the next-generation commercial cryptographic algorithm standardization, I acknowledge and agree that if my submitted algorithm is selected for the finalists, the program organizer may modify the algorithm and its implementations out of security, availability or other considerations. If my submitted algorithm is selected for the finalists and to be standardized, I commit to place no restriction on the worldwide disclosure and free adoption of algorithm and its implementations.

4. I commit that the algorithm and its implementations are my (or my team's) own original work. The implementations of my submitted algorithm ☐do/☒do not adopt code or code libraries requiring authorization by third parties. Except those clearly identified, citations and acknowledgment, all the opinions, words, diagrams and data in the submission are my own (or my team's) original research results, and the submission does not contain any published work from other individual or team.

5. *(Please check one of the following two options.)*

☒ I do not hold and do not intend to hold any patent or patent application which may cover my submitted algorithm and its implementations, and to the best of my knowledge, no patent or patent application may cover my submitted algorithm and its implementations.

☐ The following patent(s) and patent application(s) may cover my submitted algorithm and its implementations. As for patent(s) and patent application(s)

## Appendix B: Intellectual Property Statements

### B.1 Statement by Each Algorithm Submitter

I, Changchun Mu, Digital Currency Institute of the People's Bank of China, to the best of my knowledge of my submitted algorithm (name of algorithm) and its implementations (reference and optimized implementations included), do hereby declare:

1. I commit that my submitted algorithm and its implementations are in compliance with the security requirements, including but not limited to the requirements that the algorithm does not contain any backdoor or defect artificially designed and its implementations do not contain any malicious code. I do hereby commit that to the best of my knowledge, my submitted algorithm and its implementations do not contain any data, information or technology that may infringe upon trade secrets, or may involve confidentiality required by the laws and regulations of China and other countries.

2. I acknowledge and agree that my submitted algorithm and its implementations will be provided to the public for evaluation, analysis and review. I agree that any interested party, including the program organizer, public reviewers, etc., can freely use the algorithm and its implementations for evaluation purpose. I acknowledge that the program organizer will select several algorithms as candidates for the next-round evaluation or the finalists after evaluation and analysis; my submitted algorithm might not be selected as a candidate for the next-round evaluation or the finalist. I acknowledge that I will not receive financial or other compensation from the program organizer for my submitted algorithm.

3. To facilitate the next-generation commercial cryptographic algorithm standardization, I acknowledge and agree that if my submitted algorithm is selected for the finalists, the program organizer may modify the algorithm and its implementations out of security, availability or other considerations. If my submitted algorithm is selected for the finalists and to be standardized, I commit to place no restriction on the worldwide disclosure and free adoption of algorithm and its implementations.

4. I commit that the algorithm and its implementations are my (or my team's) own original work. The implementations of my submitted algorithm do/ ☒ do not adopt code or code libraries requiring authorization by third parties. Except those clearly identified, citations and acknowledgment, all the opinions, words, diagrams and data in the submission are my own (or my team's) original research results, and the submission does not contain any published work from other individual or team.

5. (Please check one of the following two options.)

☒ I do not hold and do not intend to hold any patent or patent application which may cover my submitted algorithm and its implementations, and to the best of my knowledge, no patent or patent application may cover my submitted algorithm and its implementations.

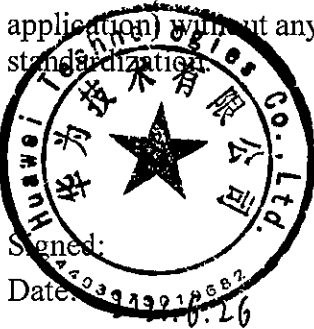
## B.2 Statement by Patent (and Patent Application) Owner(s)

*(Please check one of the following two options.)*

☒ I, Huawei Technologies Co., Ltd., No. 156, Beiqing Road, Haidian District,  
Beijing 100095, China, am the owner of the patent/patent application 202411158536.6 ;

I, (full name), (full postal address), am the authorized representative of  
the owner (full name) of the patent/patent application, (patent number/patent  
application number),

and do hereby agree to grant to any interested party, including the program organizer,  
public reviewers, etc., the right to freely use the above patent (and patent application)  
for evaluation purpose, and commit to grant to any interested party, including  
standard drafter, standard organization department, standard users, etc., a public,  
irrevocable, nonexclusive, royalty-free license for using the patent (and patent  
application) without any conditions, if the algorithm Scloud+ is selected for  
standardization.



*(Note: If there are multiple owners of the patent (and patent application), the  
statements shall be signed by each and every one of them or their authorized  
representatives. The authorized representative shall submit the letter of authorization.)*

## B.2 Statement by Patent (and Patent Application) Owner(s)

*(Please check one of the following two options.)*

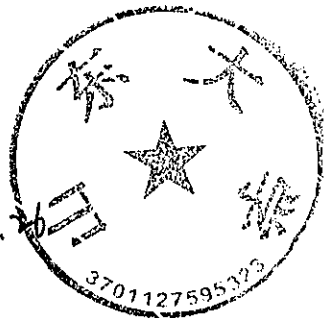
✓ I, Shandong University, No. 27, Shanda South Road, Jinan, Shandong, am the owner of the patent/patent application 202411158536.6 ;

☐ I, (full name), (full postal address), am the authorized representative of the owner (full name) of the patent/patent application , (patent number/patent application number)

and do hereby agree to grant to any interested party, including the program organizer, public reviewers, etc., the right to freely use the above patent (and patent application) for evaluation purpose, and commit to grant to any interested party, including standard drafter, standard organization department, standard users, etc., a public, irrevocable, nonexclusive, royalty-free license for using the patent (and patent application) without any conditions, if the algorithm ScIoud+ is selected for standardization.

Signed:

Date: 2026.6.26



*(Note: If there are multiple owners of the patent (and patent application), the statements shall be signed by each and every one of them or their authorized representatives. The authorized representative shall submit the letter of authorization.)*

### B.3 Statement by Reference and Optimized Implementations' Owner(s)

(Please check one of the following two options.)

☒ I, Anyu Wang, Tsinghua University, Haidian District, Beijing, China, am the owner of the submitted reference implementation and optimized implementations of the algorithm Scloud+,

☐ I, (full name), (full postal address), am the authorized representative of the owner (full name) of the submitted reference implementation and optimized implementations of the algorithm (name of algorithm), and do hereby agree that any interested party, including the program organizer, public reviewers, etc., can freely use such implementations for evaluation purpose, and commit to place no restriction on worldwide disclosure and free adoption if the algorithm is selected for standardization.

Signed: 王安宇

Date: 2026-6-26

*(Note: If there are multiple owners of the reference and optimized implementations, the statements shall be signed by each and every one of them or their authorized representatives. The authorized representative shall submit the letter of authorization.)*